

WEEK 3 Assignment

Objective:

Use Subqueries, CTEs, and Window Functions to analyze sales data from the Superstore dataset.

Tasks

Step 1: Setup Data

1. Import the Superstore dataset into a table named **superstore_raw**.
2. Create these 3 tables from it:
 - a. customers
 - b. orders
 - c. products
3. Insert data into these tables using **SELECT DISTINCT**.

Query:

Setup Database-

```
-- Create Database
```

```
CREATE DATABASE superstore_db;
```

```
USE superstore_db;
```

```
-- Verify Database
```

```
SHOW DATABASES;
```

-- Create Customers Table-

```
CREATE TABLE customers AS
```

```
SELECT DISTINCT
```

```
`Customer ID` AS customer_id,
```

```
`Customer Name` AS customer_name,
```

```
Segment,
```

Country,

City,

State,

Region

FROM superstore_raw;

SELECT *FROM customers;

customer_id	customer_name	Segment	Country	City	State	Region
CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	South
DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	West
SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	South
IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	West
HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central
PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	Central
AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	West
ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	Central
SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	East
EB-13870	Emily Burns	Consumer	United States	Orem	Utah	West
EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	West
TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	East
MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	Central
GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	Central
SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	Central

-- Create Products Table

CREATE TABLE products AS

SELECT DISTINCT

`Product ID` AS product_id,

Category, `Sub-Category` AS sub_category,

`Product Name` AS product_name

FROM superstore_raw;

SELECT *FROM products;

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	Fetch rows:
product_id	Category	sub_category	product_name	
FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	
FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	
OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	
FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	
OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	
FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	
OFF-AR-10002833	Office Supplies	Art	Newell 322	
TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	
OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	
OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	
FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	
TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	
OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	
OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	
OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner...	
OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders	
OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 3...	
OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer	
OFF-AR-10003056	Office Supplies	Art	Newell 341	
TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone	

-- Create Orders Table

CREATE TABLE orders AS

SELECT DISTINCT

`Row ID` AS row_id,

`Order ID` AS order_id,

`Order Date` AS order_date,

`Ship Date` AS ship_date,

`Ship Mode` AS ship_mode,

`Customer ID` AS customer_id,

`Product ID` AS product_id,

Sales,

Quantity,

Discount,

Profit

FROM superstore_raw;

SELECT *FROM orders;

row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	Sales	Quantity	Discount	Profit
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0	41.9136
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	OFF-LA-10000240	14.62	2	0	6.8714
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	OFF-ST-10000760	22.368	2	0.2	2.5164
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-FU-10001487	48.86	7	0	14.1694
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	OFF-AR-10002833	7.28	4	0	1.9656
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	OFF-BI-10003910	18.504	3	0.2	5.7825
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	OFF-AP-10002892	114.9	5	0	34.47
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002033	911.424	4	0.2	68.3568
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	OFF-PA-10002365	15.552	3	0.2	5.4432
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	OFF-BI-10003656	407.976	3	0.2	132.5922
15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	OFF-AP-10002311	68.81	5	0.8	-123.858
16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	OFF-BI-10000756	2.544	3	0.8	-3.816
17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0	13.3176
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	OFF-ST-10000107	55.5	2	0	9.99
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	OFF-AR-10003056	8.56	2	0	2.4824
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	TEC-PH-10001949	213.48	3	0.2	16.011

Perform Required Queries

1. Find all orders where sales are greater than the average sales.

SELECT * FROM orders WHERE sales > (SELECT AVG(sales) FROM orders);

row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	Sales	Quantity	Discount	Profit
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0	41.9136
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002033	911.424	4	0.2	68.3568
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	OFF-BI-10003656	407.976	3	0.2	132.5922
17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0	13.3176
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	240.2649
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	-1665.0522
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	123.4737
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	-46.9754
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	TEC-PH-10000486	371.168	4	0.2	41.7564
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0	298.6855
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1	7.098
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2	111.3024
73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	FUR-CH-10000513	831.936	8	0.2	-114.3912
86	US-2017-140088	5/28/2017	5/30/2017	Second Class	PO-18865	FUR-CH-10000863	301.96	2	0	33.2156
104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	TEC-AC-10001552	238.896	6	0.2	-26.8758
110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	TEC-AC-10000844	339.96	5	0.2	67.992

2. Find the highest sales order for each customer. (Subquery)

```
SELECT customer_id,
       order_id,
       sales
FROM orders o1
WHERE sales = ( SELECT MAX(sales)
                FROM orders o2
                WHERE o1.customer_id = o2.customer_id
```

);

Result Grid			
Filter Rows:			
	customer_id	order_id	sales
▶	CG-12520	CA-2016-152156	731.94
	SO-20335	US-2015-108966	957.5775
	BH-11710	CA-2014-115812	1706.184
	EB-13870	CA-2015-106320	1044.63
	TB-21520	US-2015-150630	3083.43
	GH-14485	CA-2016-117590	1097.544
	SN-20710	CA-2015-117415	532.3992
	JM-15265	CA-2016-105816	1029.95
	TB-21055	CA-2016-111682	319.41
	BS-11590	CA-2014-106376	1113.024
	PA-19060	CA-2015-129476	339.96
	HA-14920	CA-2015-110744	671.93
	JE-16165	CA-2016-114489	1951.84
	BM-11140	CA-2014-139892	8159.952
	CS-12130	CA-2014-118962	340.144
	DK-13150	CA-2014-166191	408.744
	MP-17470	CA-2015-102281	899.136
	JH-15910	CA-2016-130162	302.376
	JB-15925	CA-2015-169397	479.988
	DS-13180	CA-2015-163055	622.45

3. Calculate total sales for each customer. (CTE)

WITH customer_sales AS (SELECT customer_id, SUM(sales) AS total_sales FROM orders GROUP BY customer_id) SELECT * FROM customer_sales;

Result Grid		
Filter Rows:		
	customer_id	total_sales
▶	CG-12520	1148.7800000000002
	DV-13045	1119.4830000000002
	SO-20335	2602.5750000000001
	BH-11710	6255.3510000000001
	AA-10480	1782.872
	IM-15070	4930.4739999999999
	HP-14815	1990.314
	PK-19075	8158.654
	AG-10270	2565.25799999999994
	ZD-21925	1493.944
	KB-16585	2742.75800000000003
	SF-20065	763.54599999999998
	EB-13870	2767.219
	EH-13945	684.171
	TB-21520	4730.628
	MA-17560	4280.6209999999999
	GH-14485	1361.24400000000001
	SN-20710	3127.95920000000007
	LC-16930	4481.162

4. Find customers whose total sales are above average. (CTE + Subquery)

WITH customer_sales AS (SELECT customer_id, SUM(sales) AS total_sales FROM orders GROUP BY customer_id) SELECT * FROM customer_sales WHERE total_sales > (SELECT AVG(total_sales) FROM customer_sales);

Result Grid		Filter Rows:
	customer_id	total_sales
▶	BH-11710	6255.3510000000001
	IM-15070	4930.4739999999999
	PK-19075	8158.654
	TB-21520	4730.628
	MA-17560	4280.6209999999999
	SN-20710	3127.95920000000007
	LC-16930	4481.162
	RA-19885	3832.31399999999994
	ES-14080	4648.516
	KM-16720	4909.4720000000001
	KD-16270	8282.358
	JE-15745	6400.4549999999999
	SC-20770	4492.6569999999999
	CS-12400	3651.85850000000007
	GM-14455	3078.62400000000003
	RB-19465	4556.5660000000002
	GZ-14470	4355.15
	JM-15250	3145.37400000000003
	CL-12565	10880.5459999999998

5. Rank all customers based on total sales. (Window Function)

WITH customer_sales AS

(

SELECT customer_id, SUM(sales)

AS total_sales

FROM orders GROUP BY customer_id

)

SELECT customer_id, total_sales, RANK() OVER(ORDER BY total_sales DESC) AS
sales_rank FROM customer_sales;

Result Grid		Filter Rows:	Export
	customer_id	total_sales	sales_rank
▶	SM-20320	25043.05	1
	TC-20980	19017.8479999999998	2
	RB-19360	15117.339	3
	TA-21385	14595.62	4
	AB-10105	14355.6109999999997	5
	SC-20095	14142.3339999999999	6
	KL-16645	14071.917	7
	HL-15040	12873.2979999999999	8
	SE-20110	12209.4380000000002	9
	CC-12370	12129.072	10
	TS-21370	11885.871	11
	GT-14710	11820.1199999999997	12

6. Assign row numbers to each order within a customer.

SELECT customer_id, order_id, sales, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY sales DESC) AS row_num FROM orders;

Result Grid				
Filter Rows: Export				
	customer_id	order_id	sales	row_num
▶	AA-10315	CA-2016-103982	3930.072	1
	AA-10315	CA-2014-128055	673.568	2
	AA-10315	CA-2016-103982	431.976	3
	AA-10315	CA-2017-147039	362.94	4
	AA-10315	CA-2014-128055	52.98	5
	AA-10315	CA-2016-103982	41.72	6
	AA-10315	CA-2015-121391	26.96	7
	AA-10315	CA-2014-138100	14.94	8
	AA-10315	CA-2014-138100	14.56	9
	AA-10315	CA-2017-147039	11.54	10
	AA-10315	CA-2016-103982	2.304	11
	AA-10375	CA-2016-131065	499.98	1

7. Display top 3 customers based on total sales.

WITH customer_sales AS

(

SELECT customer_id,

SUM(sales) AS total_sales

FROM orders

GROUP BY customer_id

)

SELECT *

FROM

(

SELECT customer_id,

total_sales,

RANK() OVER(ORDER BY total_sales DESC) AS sales_rank

FROM customer_sales

) ranked_customers

WHERE sales_rank <= 3;

Result Grid			
Filter Rows:		Export:	
	customer_id	total_sales	sales_rank
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3

Final Combined Query

Command Used

```

WITH customer_sales AS
(
    SELECT c.customer_id,
           c.customer_name,
           SUM(o.sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id,
             c.customer_name
)
SELECT customer_name,
       total_sales,
       RANK() OVER(ORDER BY
total_sales DESC) AS sales_rank
FROM customer_sales;

```

Result Grid			
Filter Rows:		Export:	
	customer_name	total_sales	sales_rank
▶	Sanjit Engle	134303.818000000006	1
	Sean Miller	125215.250000000006	2
	Clay Ludtke	119686.006000000004	3
	Adrian Barton	114844.888000000012	4
	Sanjit Chand	113138.671999999998	5
	Ken Lonsdale	112575.336000000011	6
	Seth Vernon	102785.814000000017	7
	Maria Etezadi	95723.262000000022	8
	Tamara Chand	95089.239999999999	9
	Laura Armstrong	94946.742000000004	10
	Anna Häberlin	94659.527999999947	11
	Greg Tran	94560.959999999995	12
	Raymond Buch	90704.033999999999	13
	Edward Hooks	90053.819999999973	14
	Caroline Jumper	88637.935999999996	15
	Zuschuss Carroll	88232.485000000013	16
	John Lee	84627.026999999999	17
	Emily Phan	82170.911999999968	18
	Pete Kriz	81586.540000000011	19
	Harry Marie	81488.448000000012	20

Mini Project: Customer Sales Insights

1. Who are the top 5 customers?

```
WITH customer_sales AS
(
    SELECT
        c.customer_id,
        c.customer_name,
        SUM(o.sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id,
    c.customer_name
)

SELECT *
FROM customer_sales
ORDER BY total_sales DESC
LIMIT 5;
```

	customer_id	customer_name	total_sales
▶	SE-20110	Sanjit Engle	134303.81800000006
	SM-20320	Sean Miller	125215.25000000006
	CL-12565	Clay Ludtke	119686.00600000004
	AB-10105	Adrian Barton	114844.88800000012
	SC-20095	Sanjit Chand	113138.67199999998

2. Who are the bottom 5 customers?

```
WITH customer_sales AS
(
    SELECT
        c.customer_id,
        c.customer_name,
        SUM(o.sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id,
    c.customer_name
)

SELECT *
FROM customer_sales
ORDER BY total_sales ASC
LIMIT 5;
```

Result Grid			
	customer_id	customer_name	total_sales
▶	LD-16855	Lela Donovan	5.304
	TS-21085	Thais Sissman	9.666
	MG-18205	Mitch Gastineau	12.32
	CJ-11875	Carl Jackson	16.52
	RS-19870	Roy Skaria	44.65599999999999

3. Which customers made only one order?

```

SELECT
    c.customer_id,
    c.customer_name,
    COUNT(DISTINCT o.order_id) AS total_orders
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(DISTINCT o.order_id) = 1;

```

Result Grid			
	customer_id	customer_name	total_orders
▶	AO-10810	Anthony O'Donnell	1
	AR-10570	Anemone Ratner	1
	CJ-11875	Carl Jackson	1
	JC-15385	Jenna Caffey	1
	JR-15700	Jocasta Rupert	1
	LD-16855	Lela Donovan	1
	MG-18205	Mitch Gastineau	1
	PH-18790	Patricia Hirasaki	1
	RE-19405	Ricardo Emerson	1
	RM-19750	Roland Murray	1
	SM-20905	Susan MacKendrick	1
	TC-21145	Theresa Coyne	1

4. Which customers have above-average sales?

```

WITH customer_sales AS
    SELECT
        c.customer_id,
        c.customer_name,
        SUM(o.sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id

```

```

GROUP BY c.customer_id,
c.customer_name
)
WHERE total_sales >
(
SELECT AVG(total_sales)
FROM customer_sales
)
SELECT *
FROM customer_sales
);

```

customer_id	customer_name	total_sales
BH-11710	Brosina Hoffman	50042.808000000005
IM-15070	Irene Maddox	34513.317999999985
PK-19075	Pete Kriz	81586.54000000011
KB-16585	Ken Black	30170.337999999974
EB-13870	Emily Burns	24904.970999999997
TB-21520	Tracy Blumstein	42575.651999999998
MA-17560	Matt Abelman	38525.588999999999
LC-16930	Linda Cazamias	35849.295999999915
RA-19885	Ruben Ausman	26826.198000000033
ES-14080	Erin Smith	37188.127999999946
LH-16900	Lena Hernandez	20657.987999999999
KM-16720	Kunst Miller	39275.775999999976

5. What is the highest order value per customer?

```

SELECT
c.customer_id,
c.customer_name,
MAX(o.sales) AS
highest_order_value
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id,
c.customer_name
ORDER BY highest_order_value
DESC;

```

customer_id	customer_name	highest_order_value
SM-20320	Sean Miller	22638.48
TC-20980	Tamara Chand	17499.95
RB-19360	Raymond Buch	13999.96
TA-21385	Tom Ashbrook	11199.968
HL-15040	Hunter Lopez	10499.97
AB-10105	Adrian Barton	9892.74
SC-20095	Sanjit Chand	9449.95
BS-11365	Bill Shonely	9099.93
SE-20110	Sanjit Engle	8749.95
CC-12370	Christopher Conant	8399.976
KL-16645	Ken Lonsdale	8187.65
BM-11140	Berky Martin	8159.952